Amazon Ecommerce Analysis

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OBJECTIVE QUESTIONS

1. What is the total number of attributes in the customer table?

A. The **Customer** table contains the following three fields:

- **CustomerID**: A unique value assigned to each customer.
- **Customer Age**: Represents the age of the customer.
- **Customer Gender**: Indicates the customer's gender, where **M** denotes male and **F** denotes female.

2. How will you get the "Customer's" ages in the "Order" tables according to customer IDs?

A. **Procedure:**

- In **Data View**, the **Orders** table was selected from the Fields pane on the right.
- From the top menu, the **Modeling** tab was opened and **New Column** was chosen.
- The following DAX expression was entered:

```
CustomerAge=LOOKUPVALUE(Customers[CustomerAge],Customers[CustomerID],Orders[CustomerID])
```

- After executing the formula, a new column named **CustomerAge** was successfully added to the **Orders** table.

3. In analyzing the dataset with Power BI, ensure data cleaning to address inconsistencies and missing values before further analysis.

A. In Power BI, the **Power Query Editor** is used to perform data cleaning and preparation tasks such as:

- Eliminating duplicate records.
- Removing rows with missing or blank values.
- Validating and correcting data types.
- Deleting unnecessary or empty columns.

4. How can we calculate the total revenue generated by all the sales?

A. To calculate **Total Revenue**, a measure was created using the following steps:

- Opened **Data View** or **Model View** in Power BI.
- Selected the **Orders** table.
- From the ribbon menu, choose **New Measure**.
- Entered the required **DAX formula** to compute total revenue.

```
Total Revenue = SUM(Orders[Sale Price])
```

107.24M
Total Revenue

5. What is the total number of unique customers who made purchases each year? Is there any increase in the number over the years?

A. Extracting the Year from Order Date

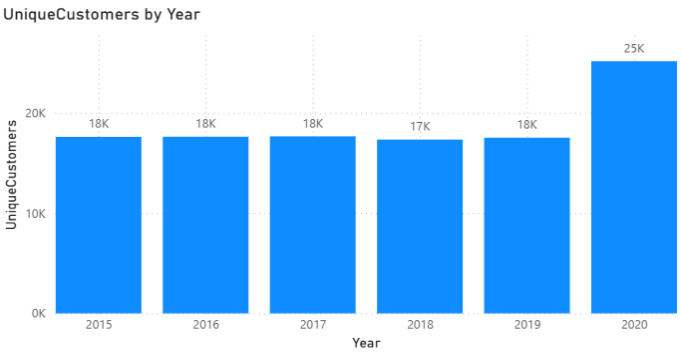
- Open **Power Query** using **Transform Data**.
- Select the **Order Date** column.
- Navigate to **Add Column** → **Date** → **Year**.
- This action generates a new column named **Year**, capturing the year in which each order was placed.

Creating a Measure for Year-wise Unique Customers

- Go to **Data View** and select the **Orders** table.
- Click on **New Measure**.
- Define the following DAX measure:

```
UniqueCustomersPerYear = DISTINCTCOUNT(Orders[CustomerID])
```

Visualization:



Observations:

- The number of unique customers shows a consistent upward trend from 2019 to 2020, which may reflect successful marketing efforts or a broader product offering.
- A minor drop in customer count between 2017 and 2018 could point to challenges such as stock issues, pricing adjustments, or heightened market competition.

6. How can we determine the total number of unique products available in the company?

A. To calculate the total count of unique products, a measure was created using the following approach:

- Navigated to **Data View** and selected the **Orders** table.
- Choose a new **Measure** from the ribbon.
- Applied the following DAX formula:

```
UniqueProducts = DISTINCTCOUNT(Orders[Product])
```

7. What is the average number of days it takes for products to be delivered, get the metric for only the delivered orders.

A. To compute the average number of days taken to deliver products, the following approach was used:

Step 1: Create a calculated column for delivery time

- Open **Data View** and select the **Orders** table.
- Add a new calculated column.

- Use the following DAX formula:

```
DeliveryDaysCount=DATEDIFF(Orders[OrderDate], Orders[Delivery Date],DAY)
```

- This column stores the total number of days between the order date and the delivery date.

Step 2: Create a measure for average delivery duration (Delivered orders only)

- Create a new measure in the Orders table.
- Apply a filter to include only records with a Delivered status:

```
AvgDeliveryDaysForDeliveredOrders=CALCULATE(AVERAGE(Orders[DeliveryDaysCount]),Orders[Status] = "Delivered")
```

- This measure calculates the average delivery time exclusively for completed orders.

9.41

AvgDeliveryTime_Delivered

8. Which products, categories, and subcategories are the most popular?

A. a) Identifying the Most Popular Products:

- Insert a **Table visual** into the report canvas.
- Add the **Product** column and the **Quantity** column to the table.
- Set the **Quantity** column to **Count** (to get the number of occurrences per product) and sort the table in **descending order** based on this count.

Product	Count of Order Quantity
10.1" Business Tablet with MT6582 Quad-Core Processor	2569
100%Cotton 4 Piece Short Sleeve T-Shirts - Multicolour	2569
6030 3.1 Bluetooth Home Theatre With Remote Control - Black + Free Smartwatch	2568
8 Cubes Plastic Wardrobe - Blue/White	2569
Aichun Beauty Eight Pack Essential Oil - 30ml	2566
Amazon Fire HD 8 Kids Tablet 32GB HDD - 2GB RAM - 8" Blue	2569
Avon Soft Musk Eau de Toilette Spray - 50ml	2569
B5 HiFi 5.0 Ture Wireless Headsets Auto Pair Touch - Black	2569
Blood Pressure Monitor Digital Wrist BP Pulse Monitor Meter Heart Rate Measure	2569
Boys Sneakers Casual Kids Sports Shoes-Gold	2569
Canon EOS 600D 18MP CMOS DSLR Camera - Black	2569
Canon EOS 60D CMOS DSLR Camera Bundle - 18 - 55mm Lens - Black	2567
Clere Avocado Milk Body Lotion With Vitamins E+A - 400ml	2569
Clere Radiance Oil Control Toner - 100ml	2569
Cq Amaigrissant Slimming Tea - 20 Tea Bags	2567
Fashion 4-Piece Leather HandBag Set - Black	2567
Fashion Boys Sneakers Children Outdoor Shoes-Black	2567
Fashion Girl's Dress Kids Children Newborn Baby Dinner Party Princess Dress Ball Gown	2566
Fashion Girls' Patent Leather Stitching Shoes - Black	2568

b) Identifying the Most Popular Categories:

- Insert a **Table visual** into the report canvas.
- Add the **Product Category** column and the **Quantity** column to the table.
- Set the **Quantity** column to **Count** (to get the number of occurrences per category) and sort the table in **descending order** based on this count.

Product Category	Count of Order Quantity
Health and beauty	35951
Fashion	33383
Phones and Tablet	17978
Home and Office	15408
Electronics	10271
Total	**112991**

c) Identifying the Most Popular Subcategories:

- Insert a **Table visual** into the report canvas.
- Add the **Subcategories** column and the **Quantity** column to the table.
- Set the **Quantity** column to **Count** (to get the number of occurrences per subcategory) and sort the table in **descending order** based on this count.

SubCategory	Count of Order Quantity
Beauty and personal care	7707
Boy's fashion	7704
Digital Cameras	5136
Fragrances	5137
Girl's fashion	7703
Home and Furniture	5137
Home Audio	5135
Kitchen and dinning	7703
Medical supplies and Equipment	10269
Men's fashion	10272
Mobile accessories	5137
Mobile phones	7703
Tablets	5138
Tools and Home Improvement	2568
Vitamins & Dietary Supplements	12838
Women's fashion	7704
Total	**112991**

9. Which products have seen an increase or decrease in sales over the year?

A. To analyze the yearly trend of product sales, the following approach was used:

Step 1: Create a measure for Previous Year Sales

- Navigate to **Data View** or **Model View**.
- Select the **Orders** table and create a **New Measure**.
- Define a measure to calculate sales from the previous year (using an appropriate DAX formula for prior year comparison).

```
Revenue_LastYear = CALCULATE(SUM(Orders[Sale Price]),
FILTER(Orders,YEAR(Orders[OrderDate])=YEAR(MAX(Orders
[OrderDate])) - 1))
```

Step 2: Create a measure for Current Year Revenue

- Purpose: Calculate total revenue for the current year based on the latest order date in the dataset.

```
Revenue_CurrentYear = SUM(Orders[Sale Price])
```

Step 3: Create a measure for Sales Difference

- Compute the difference between current year sales and previous year sales for each product.

```
Revenue_Change=[Revenue_CurrentYear]-[Revenue_LastYea
r]
```

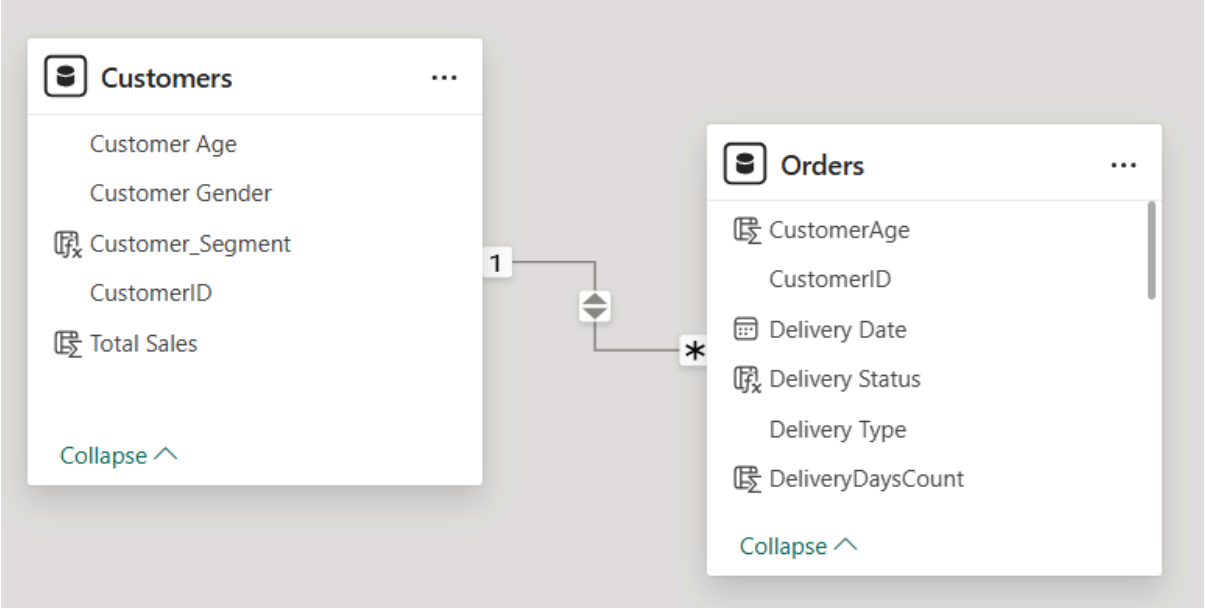
Product	Revenue_CurrentYear	Revenue_LastYear	Revenue_Change
10.1" Business Tablet with MT6582 Quad-Core Processor	5293898	732426	4561472
100%Cotton 4 Piece Short Sleeve T-Shirts - Multicolour	1160562	186894	973668
6030 3.1 Bluetooth Home Theatre With Remote Control - Black + Free Smartwatch	5405068	806046	4599022
8 Cubes Plastic Wardrobe - Blue/White	2513287	393573	2119714
Aichun Beauty Eight Pack Essential Oil - 30ml	513528	79115	434413
Amazon Fire HD 8 Kids Tablet 32GB HDD - 2GB RAM - 8" Blue	11236628	1882052	9354576
Avon Soft Musk Eau de Toilette Spray - 50ml	1113510	161215	952295
B5 HiFi 5.0 Ture Wireless Headsets Auto Pair Touch - Black	774909	121152	653757
Blood Pressure Monitor Digital Wrist BP Pulse Monitor Meter Heart Rate Measure	933963	148178	785785
Boys Sneakers Casual Kids Sports Shoes-Gold	1745896	256549	1489347
Canon EOS 600D 18MP CMOS DSLR Camera - Black	13890491	2082643	11807848
Canon EOS 60D CMOS DSLR Camera Bundle - 18 - 55mm Lens - Black	12094384	1964631	10129753

10. While modeling the data relationships, what will be the type of relationship between the customer ID of Orders and customer tables?

A. In **Model View**, drag the **CustomerID** column from the **Customers** table and connect it to the **CustomerID** column in the **Orders** table.

The relationship created is a **one-to-many** relationship:

- **One** customer in the **Customers** table can have **many orders** in the **Orders** table.



11. How have you handled the null values in the data?

A. Null and blank values were managed using a combination of **Power Query transformations** and **DAX measures**, depending on the requirement.

Handling null values in Power Query:

- Open **Power BI Desktop** and navigate to **Transform Data** to access Power Query.
- Select the relevant column where null values need to be addressed.
- Use the column filter to exclude **null** entries, which removes rows containing missing values.

Managing blank values using DAX:

- In **Report or Model View**, create a new measure from the **Modeling** tab.

- Apply logic to replace blank values with a default value to avoid calculation issues.
  Example:

```
SalePrice_Adjusted = IF(ISBLANK(Orders[Sale Price]), 0, Orders[Sale Price])
```

Removing fully blank rows:

- In Power Query, review the dataset for rows where all fields are empty.
- Use **Remove Rows** → **Remove Blank Rows** to eliminate such records from the dataset.

12. Were there any data format issues in the data, and if there were/are how you would handle them?

A. In this dataset, there were no data type inconsistencies that required changes. All fields were already in the appropriate format for analysis.

However, if any data format issues were identified, they could be resolved using the **Power Query Editor** by reviewing and transforming the column data types as needed to ensure accuracy and consistency during analysis.

13. When we add a column in Power Query what's the code that comes in M language in the formula bar? What do you know about M-query?

A. **M Query** is the language used in Power Query to clean and transform data before it is loaded into the Power BI model. It follows a functional approach to apply step-by-step transformations.

Whenever a column is created or modified in Power Query, the tool automatically produces the corresponding **M code** in the formula bar, ensuring that every transformation is traceable and repeatable.

M Query operates during the **ETL phase (Extract, Transform, Load)**, making it ideal for data cleansing, reshaping, and enrichment prior to analysis.

Example: M Code for Creating a Custom Column

```
= Table.AddColumn(LastStep, "AdjustedAmount", each [Revenue] * 1.1, type number)
```

Explanation:

Table.AddColumn: Adds a new calculated column to the dataset.

LastStep: References the previous step in the query workflow.

"AdjustedAmount": Name assigned to the newly created column.

each [Revenue] * 1.1: Applies row-level logic to calculate updated values.

type number: Defines the data type for the new column.

14. Identify the top 5 most valuable customers using a composite score that combines three key metrics: (SQL)
a. Total Revenue (50% weight): The total amount of money spent by the customer.
b. Order Frequency (30% weight): The number of orders placed by the customer, indicating their loyalty and engagement.
c. Average Order Value (20% weight): The average value of each order placed by the customer, reflecting the typical transaction size.

A. The Composite Score will be calculated as:

(0.5×Total Revenue)+(0.3×Order Frequency)+(0.2×Average Order Value)

Query :

```sql
WITH customer_metrics AS (
    SELECT
        cust.CustomerID,
        SUM(ord.SalePrice) AS revenue_sum,
        COUNT(ord.OrderID) AS order_count,
        AVG(ord.SalePrice) AS avg_order_value
    FROM Customers AS cust
    INNER JOIN Orderss AS ord
        ON cust.CustomerID = ord.CustomerID
    GROUP BY cust.CustomerID
)
SELECT
    CustomerID,
    revenue_sum AS TotalRevenue,
    order_count AS OrderFrequency,
    avg_order_value AS AverageOrderValue,
    (revenue_sum * 0.5
     + order_count * 0.3
     + avg_order_value * 0.2) AS CompositeScore
FROM customer_metrics
ORDER BY CompositeScore DESC
LIMIT 5;
```

Output :

CustomerID	TotalRevenue	OrderFrequency	AverageOrderValue	CompositeScore
230537641	8180.00	1	8180.000000	5726.3000000
230502503	8180.00	1	8180.000000	5726.3000000
230526514	8180.00	1	8180.000000	5726.3000000
230527136	8180.00	1	8180.000000	5726.3000000
230544339	8180.00	1	8180.000000	5726.3000000

15. Calculate the month-over-month growth rate in total revenue across the entire dataset. (SQL)

A. Query :

```sql
WITH monthly_sales_summary AS (
    SELECT
        YEAR(OrderDate) AS sales_year,
        MONTH(OrderDate) AS sales_month,
        SUM(SalePrice) AS monthly_revenue
    FROM Orderss
    GROUP BY sales_year, sales_month
)
SELECT
    sales_year AS Year,
    sales_month AS Month,
    monthly_revenue AS TotalRevenue,
    (
        (monthly_revenue
         - LAG(monthly_revenue) OVER (ORDER BY sales_year, sales_month))
        / LAG(monthly_revenue) OVER (ORDER BY sales_year, sales_month)
    ) * 100 AS MoMGrowthRate
FROM monthly_sales_summary
ORDER BY sales_year, sales_month;
```

Output :

	Year	Month	TotalRevenue	MoMGrowthRate
	2015	6	1378115.15	-9.172163
	2015	7	1369651.25	-0.614165
	2015	8	1520687.45	11.027347
	2015	9	1281755.35	-15.712111
	2015	10	1507927.60	17.645509
	2015	11	1398724.10	-7.241959
	2015	12	1465184.55	4.751505
	2016	1	1276295.55	-12.891823
	2016	2	1317497.25	3.228226

16. Calculate the rolling 3-month average revenue for each product category. (SQL)

A. Query :

```sql
WITH category_monthly_sales AS (
    SELECT
        YEAR(OrderDate) AS sales_year,
        MONTH(OrderDate) AS sales_month,
        ProductCategory,
        SUM(SalePrice) AS monthly_revenue
    FROM Orderss
    GROUP BY sales_year, sales_month, ProductCategory
)
SELECT
    sales_year AS Year,
    sales_month AS Month,
    ProductCategory,
    AVG(monthly_revenue) OVER (
        PARTITION BY ProductCategory
        ORDER BY sales_year, sales_month
        ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
    ) AS Rolling3MonthAvgRevenue
FROM category_monthly_sales
ORDER BY sales_year, sales_month;
```

Output :

Year	Month	ProductCategory	Rolling3MonthAvgRevenue
2015	1	Electronics	527626.250000
2015	1	Fashion	156522.000000
2015	1	Health and beauty	150503.600000
2015	1	Home and Office	130176.000000
2015	1	Phones and Tablet	519549.200000
2015	2	Electronics	506260.750000
2015	2	Fashion	149063.500000
2015	2	Health and beauty	149854.800000

17. Update the orders table to apply a 15% discount on the `Sale Price` for orders placed by customers who have made at least 10 orders. (SQL)

A. Query :

```sql
UPDATE orderss
SET SalePrice = SalePrice * 0.85
WHERE CustomerID IN (
    SELECT CustomerID
    FROM orderss
    GROUP BY CustomerID
    HAVING COUNT(OrderID) >= 10
);
```

18. Calculate the average number of days between consecutive orders for customers who have placed at least five orders. (SQL)

A. Query :

```sql
WITH ordered_customer_purchases AS (
    SELECT
        o.CustomerID,
        o.OrderID,
        o.OrderDate,
        LEAD(o.OrderDate) OVER (
            PARTITION BY o.CustomerID
            ORDER BY o.OrderDate
        ) AS following_order_date
    FROM Orderss o
)
SELECT
    CustomerID,
    AVG(
        DATEDIFF(following_order_date, OrderDate)
    ) AS AvgDaysBetweenOrders
FROM ordered_customer_purchases
WHERE following_order_date IS NOT NULL
GROUP BY CustomerID
HAVING COUNT(*) >= 5;
```

Output :

	CustomerID	AvgDaysBetweenOrders

19. Identify customers who have generated revenue that is more than 30% higher than the average revenue per customer. (SQL)

A. Query :

```sql
WITH customer_revenue_summary AS (
    SELECT
        CustomerID,
        SUM(SalePrice) AS customer_total_revenue
    FROM Orderss
    GROUP BY CustomerID
),
overall_avg_revenue AS (
    SELECT
        AVG(SalePrice) AS overall_avg_sale
    FROM Orderss
)
SELECT
    cr.CustomerID,
    cr.customer_total_revenue AS TotalRevenue
FROM customer_revenue_summary cr
JOIN overall_avg_revenue oar
    ON cr.customer_total_revenue > oar.overall_avg_sale * 1.30;
```

Output :

CustomerID	TotalRevenue
230530055	4406.00
230560487	1935.00
230565789	4408.00
230567063	4717.00
230540401	4708.00
230494227	8165.00
230562223	3112.00
230475583	8178.00
230479611	1941.00

20. Determine the top 3 product categories that have shown the highest increase in sales over the past year compared to the previous year. (SQL)

A. Query :

```sql
WITH yearly_category_sales AS (
    SELECT
        ProductCategory,
        YEAR(OrderDate) AS sales_year,
        SUM(SalePrice) AS year_total_sales
    FROM Orderss
    WHERE YEAR(OrderDate) IN (2019, 2020)
    GROUP BY ProductCategory, sales_year
)
SELECT
    curr.ProductCategory,
    curr.year_total_sales AS CurrentYearSales,
    prev.year_total_sales AS PreviousYearSales,
    (curr.year_total_sales - prev.year_total_sales) AS SalesIncrease
FROM yearly_category_sales curr
JOIN yearly_category_sales prev
    ON curr.ProductCategory = prev.ProductCategory
   AND curr.sales_year = 2020
   AND prev.sales_year = 2019
ORDER BY SalesIncrease DESC
LIMIT 3;
```

Output :

	ProductCategory	CurrentYearSales	PreviousYearSales	SalesIncrease
▶	Phones and Tablet	8544101.20	6026542.20	2517559.00
	Electronics	7475906.50	5075494.25	2400412.25
	Fashion	2764786.00	1948038.00	816748.00